Introduction to Software Engineering

Software Requirements Specification

The student team is required to complete the **Software Requirements Specification** documentation for the assigned course project, following the attached template.



Software Engineering Department
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Software Requirements Specification

Objectives

This document focus on the following topics:

- ✔ Completing the Software Requirements Specification (SRS) document with the following sections:
  - Problem Statement
  - Stakeholders
  - Functional, non-functional and domain requirements
  - Use case model and use case specifications
  - Prototype / mockups
  - Traceability and requirements review
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Software Requirements Specification (SRS) document.

1 Member Contribution Assessment

ID	Name	Role(s)	Contribution (%)	Evidence
24127462	Huỳnh Trung Ngạn	PM	33%	https://github.com/nelan138/vible-online-teaching-platform/issues/47 https://github.com/nelan138/vible-online-teaching-platform/issues/52
24127175	Lê Đoàn Nhật Huy	BA		
24127445	Trương Hiển Long	Dev		
24127053	Trần Hoàng Minh Khang	Designer		

2 Introduction

2.1 Purpose and Scope

This Software Requirements Specification (SRS) defines the requirements for Vible, a web-based online teaching platform for teachers and students. It establishes the expected system behaviour, quality requirements, business rules, use cases, and acceptance criteria as a basis for design, implementation, testing, and requirements review.

Vible brings course activities into one place. The core scope includes account registration and login, role-based access, course management, enrollment and roster management, course material publishing and access, and assignment creation and submission (F01–F05). Grading and feedback and course announcements and discussions are Should-have features (F06–F07); notifications and profile customization are Could-have features (F08–F09).

Out of scope are real-time private one-to-one messaging, payment processing, live video conferencing, institution-wide user directory synchronization, and bulk administrative roster imports, as defined in PA1.

The intended readers are the project team, including the project manager, business analyst, designer, developers, and testers; the course instructor and teaching assistants; and teacher and student representatives involved in reviewing the requirements.

2.2 Glossary

Term	Definition
SRS	Software Requirements Specification; the document defining the system's required behaviour and quality constraints.
LMS	Learning Management System; a platform for managing courses, learning materials, assignments, and related activities.
Teacher	A user who creates and manages their own courses, materials, participants, and assignments.
Student	A user who requests course enrollment, accesses enrolled course content, and submits assignments.

Roster	The list of enrolled students and pending enrollment requests for a course.
Invite code / link	An invitation used by a student to request enrollment in a course; enrollment requires teacher approval.
Course material	Learning content published by a teacher for enrolled students to view or download.
Submission	Work a student submits for an assignment; work submitted after its due date is marked late.
MoSCoW	Requirement priorities: Must-have, Should-have, Could-have, and Won't-have for this delivery.
FR / NFR / BR / UC	Identifiers for functional requirements, non-functional requirements, business rules, and use cases, respectively.

3 Problem Statement

Teachers and students may use chat apps such as Zalo or Discord, spreadsheets, and email to manage classroom activities. When materials, assignments, and messages are spread across these tools, students can struggle to find the information they need, while teachers have to track participants and submissions in separate places.

Vible aims to bring these activities into one web platform. Teachers can manage their courses, enroll students, publish materials, and create assignments with deadlines. Students can access their enrolled courses and submit their work. The goal is to make everyday course activities easier to organize and follow.

The platform will run in modern web browsers on desktop and mobile devices with an internet connection. Following PA1, development uses Nuxt with TypeScript, PostgreSQL, and Cloudflare hosting. Work is limited by the course schedule and the planned use of free services, so Must-have features take priority.

4 Stakeholders

#	Stakeholder	Description	Interests / concerns
1	Teachers / Instructors	End users who create and manage courses, materials, participants, and assignments.	An easy way to organize course activities, manage enrollment, and review student work.
2	Students	End users who join courses, access learning materials, and submit assignments.	Easy access to course content, clear deadlines, and reliable submission of their work.
3	Project team	Plans, analyzes, designs, implements, and tests Vible.	Clear requirements, manageable scope, and delivery of the Must-have features within the course schedule.
4	Course instructor and teaching assistants	Review project deliverables and assess the team's work.	Requirements that meet the assignment criteria, traceable deliverables, and evidence of teamwork.

5 Requirements

5.1 Functional Requirements

ID	Requirement	Feature	Priority	Source
FR-01A	The system shall allow users to register new accounts.	F01	Must	PA1 Section 2.4, F01.
FR-01B	The system shall allow users to log in using credentials.		Must	PA1 Section 2.4, F01.
FR-01C	The system shall enforce role-based access for every user (Student or Teacher).		Must	PA1 Section 2.4, F01.
FR-02A	The system shall allow teachers to create new courses.	F02	Must	PA1 Section 2.4, F02.
FR-02B	The system shall allow teachers to update their courses (rename, delete,...).		Must	PA1 Section 2.4, F02.
FR-03A	The system shall allow teachers to enroll students via invite code (or link).	F03	Must	PA1 Section 2.4, F03. Invite-code/link flow: PA2 working draft; team confirmation pending.
FR-03B	The system shall allow teachers to remove students from courses.		Must	PA1 Section 2.4, F03.
FR-03C	The system shall allow students to join a course via invite code or link.		Must	PA1 Section 2.4, F03. Invite-code/link flow: PA2 working draft; team confirmation pending.
FR-04A	The system shall allow teachers to update course materials (publish, delete,...)	F04	Must	PA1 Section 2.4, F04.

FR-04B	The system shall allow enrolled students to view / download published course materials.		Must	PA1 Section 2.4, F04.
FR-05A	The system shall allow teachers to create in-app assignments with a due date.	F05	Must	PA1 Section 2.4, F05.
FR-05B	The system shall allow students to submit assignments.		Must	PA1 Section 2.4, F05.
FR-05C	The system shall flag late submissions.		Must	PA1 Section 2.4, F05. Late-submission handling: PA2 working draft; policy confirmation pending.
FR-06A	The system shall allow teachers to grade and give feedback to student submissions.	F06	Should	PA1 Section 2.4, F06.
FR-07A	The system shall provide a dedicated channel for each course where teachers post announcements for students who are enrolled in that course.	F07	Should	PA1 Section 2.4, F07.
FR-07B	The system shall provide a dedicated channel for each course where teachers and students who are enrolled in that course can exchange messages.		Should	PA1 Section 2.4, F07.
FR-08A	The system shall send notifications to users via a dedicated channel or web page.	F08	Could	PA1 Section 2.4, F08.
FR-08B	The system shall send notifications to users via registered emails.		Could	PA1 Section 2.4, F08.
FR-09A	The system shall allow users to update their profile details (display name, avatar, bio).	F09	Could	PA1 Section 2.4, F09.
FR-09B	The system shall allow users to view other users' profiles.		Could	PA2 working draft, FR-09B. Viewing other profiles extends PA1 F09; scope and visible fields require team confirmation.

5.2 Non-Functional Requirements

ID	Category	Requirement (measurable)	How it will be verified
NFR-01	Performance	The system shall return authenticated course-list, material-list, and assignment-list API responses within 2 seconds at P95, with fewer than 1% unexpected server errors, during a 10-minute run after a 1-minute warm-up. Proposed workload: 50 concurrent users, 20 courses, 500 enrollments, 200 material records, and 200 assignments; one request per user every 5 seconds, evenly distributed across the three endpoints. File transfer time is excluded.	Run a load test against the deployment candidate from a fixed test runner; record its location, network latency, dataset, P95 by endpoint, and server-error rate.
NFR-02	Security - authorization	The system shall enforce authentication, role, course ownership, and enrollment checks on the server for all protected Must-have operations. Every unauthorized request in the access-control test matrix shall disclose no protected course/submission data and cause no mutation. Hiding UI controls alone shall not grant or restrict access.	Exercise APIs and file-download access as an unauthenticated user, enrolled student, non-enrolled student, course owner, and another teacher. All prohibited combinations must be denied; verify stored data is unchanged.
NFR-03	Usability	At least 4 of 5 first-time student test participants shall independently join a course, locate its material, and submit a valid assignment file within 5 minutes in total using a provided account, valid invitation, and prepared file, excluding sign-in time.	Run a moderated task-based test with the same instructions and dataset; record completion time, success, and any assistance. This is a proposed test, not a completed user study.
NFR-04	Accessibility	All controls needed for registration, login, joining a course, accessing materials, and submitting an assignment shall be operable by keyboard, with a visible focus indicator. Each input shall have a programmatic label, and validation errors shall identify the affected input.	Perform the five workflows using keyboard only; inspect labels and error associations using browser accessibility tools. No inaccessible control may block a Must-have workflow.
NFR-05	Compatibility and layout	All Must-have acceptance scenarios shall pass on the stable versions of Chrome, Edge, and Firefox available at the test baseline. At viewport widths of 360 and 1366 CSS pixels, primary controls shall remain reachable and readable; wide tables may scroll within their own container.	Record exact browser versions and run the scenarios at both widths. Inspect for clipped controls, page-level horizontal scrolling, and overlapping text.

NFR-06	Reliability and data integrity	The system shall report a submission as successful only after its file and submission record are persistently stored. Across 10 injected interruption/failure trials during upload or save, no incomplete submission shall be shown as successful and any previously accepted submission shall remain accessible.	Interrupt uploads and simulate storage/database failures in a test environment; inspect the user message, stored file, submission record, and previous accepted submission after each trial.
NFR-07	Security - transport and credentials	The deployed application shall use HTTPS for authentication and protected data/file access. For credential-based authentication, passwords shall not appear in plaintext in persistent storage, application logs, or API responses.	Inspect deployed requests, a test-account database record, application logs, and API responses; verify secure transport and review the password-storage implementation.
NFR-08	Upload validation	The system shall enforce BR-06 file type and size limits on the server for material and assignment uploads. It shall reject every invalid file in the boundary/type test set with a clear error and shall not publish the rejected file as a material or successful submission.	Test an allowed file, an empty file, exactly 20 MiB, 20 MiB plus one byte, a disallowed type, and a file whose extension conflicts with its validated content type.

5.3 Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	Each user has exactly one role: Student or Teacher.	FR-01C
BR-02	A teacher can create and manage many courses, but each course belongs to only one teacher.	FR-02A, FR-02B
BR-03	A student can join many courses via invite code or link, and a course can have many students without duplicates.	FR-03A, FR-03C
BR-04	Only the course teacher and enrolled students can access the course's materials, assignments, announcements, and discussions.	FR-03B, NFR-02
BR-05	A course can have many materials, and enrolled students can only view or download published materials.	FR-04A, FR-04B
BR-06	Uploaded files (materials and submissions) must not be empty, must be at most 20 MiB, and must have an allowed file type.	FR-04A, FR-05B, NFR-08

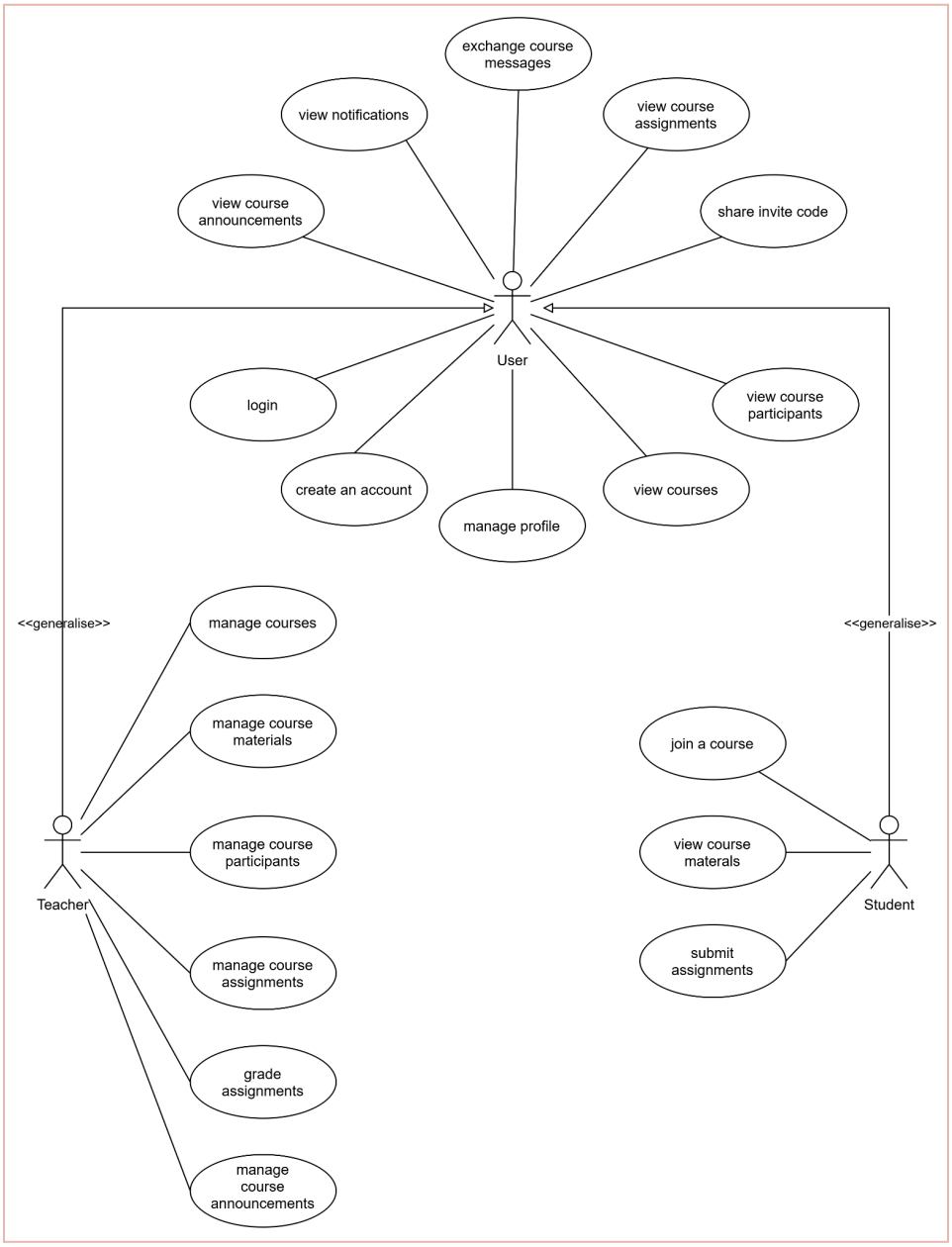
BR-07	Each assignment must have a due date, and any submission made after the due date is flagged as late.	FR-05A, FR-05B, FR-05C
BR-08	Only the course teacher can grade submissions and post announcements, while both the teacher and students can chat in discussions.	FR-06A, FR-07A, FR-07B

5.4 Acceptance Criteria

Feature	Given	When	Then form
F01	A user is not logged in.	They submit valid credentials.	The system authenticates them and authorizes access according to their assigned role.
F02	A teacher is logged in.	They create a course with valid details, or update or delete a course they own.	The system creates, updates, or deletes the course as requested.
F03	A logged-in student has a valid course invite code or link.	They use it to join the course.	The system enrolls them and adds them to the course's roster after the teacher of that course confirms.
F04	A teacher owns a course.	They publish valid course material.	The system makes the material available for enrolled students to view or download.
F05	A student is enrolled in a course with an assignment and a deadline.	They submit a valid file for the assignment.	The system saves the submission and flags it as late if submitted after the deadline.

6 Use Case Model

6.1 Use Case Diagram



6.2 Actors

Actor	Description
Teachers	Creates and manages courses, materials, participants, assignments, and announcements; grades submissions and participates in discussions.
Students	Joins courses, accesses materials and announcements, submits assignments, views grades and feedback, and participates in discussions.

6.3 Use Case List

ID	Use case	Primary actor	Priority	Related FR
UC-01	Create an account	Student / Teacher	Must	FR-01A
UC-02	Log in	Student / Teacher	Must	FR-01B
UC-03	View courses	Student / Teacher	Must	FR-02A, FR-02B, FR-03C
UC-04	Manage courses	Teacher	Must	FR-02A, FR-02B
UC-05	Manage course materials	Teacher	Must	FR-04A
UC-06	Join a course	Student	Must	FR-03C
UC-07	View course materials	Student	Must	FR-04B
UC-08	View course participants	Teacher / Student	Must	FR-03A, FR-03B, FR-03C
UC-09	Share invite code	Teacher / Student	Must	FR-03A, FR-03C
UC-10	Manage course participants	Teacher	Must	FR-03A, FR-03B
UC-11	View course assignments	Teacher / Student	Must	FR-05A, FR-05B
UC-12	Manage course assignments	Teacher	Must	FR-05A
UC-13	Submit an assignment	Student	Must	FR-05B, FR-05C
UC-14	Grade assignments	Teacher	Should	FR-06A

UC-15	Manage course announcements	Teacher	Should	FR-07A
UC-16	Exchange course messages	Teacher / Student	Should	FR-07B
UC-17	Manage profile	Student / Teacher	Could	FR-09A
UC-18	View notifications	Student / Teacher	Could	FR-08A
UC-19	View course announcements	Student / Teacher	Could	FR-07A

7 Use Case Specifications

7.1 Use Case UC-01

Use case ID	UC-01
Use case name	Create an account
Brief description	A user registers as a Student or Teacher.
Primary actor	Student / Teacher
Secondary actors	None
Priority	Must
Related requirements	FR-01A, FR-01C
Trigger	The user selects registration.
Pre-conditions	The user is not logged in.
Post-conditions	An account is created with the supplied details and selected role.
Main flow	1. The system displays the registration form. 2. The user enters their display name, email, password, and role, then submits. 3. The system validates the details and checks that the email is not already registered. 4. The system creates the account and confirms successful registration.
Alternative flows	A1 (step 2): The user cancels; no account is created.
Exception flows	E1 (step 3): Details are missing or invalid; the system identifies the errors and allows correction. E2 (step 3): The email is already registered; the system informs the user and allows correction. E3 (step 4): Saving fails; the system reports that registration was unsuccessful.
Business rules	Each account has a unique email and exactly one role: Student or Teacher.
Non-functional constraints	Use HTTPS; never store plaintext passwords. Support keyboard navigation, labeled fields, and identifiable validation errors.

7.2 Use Case UC-02

Use case ID	UC-02
Use case name	Log in
Brief description	A user logs in to access the platform.
Primary actor	Student / Teacher
Secondary actors	None
Priority	Must
Related requirements	FR-01B, FR-01C
Trigger	The user selects login.
Pre-conditions	The user has an account and is not logged in.
Post-conditions	The user is authenticated and authorized according to their assigned role.
Main flow	1. The system displays the login form. 2. The user enters their email and password, then submits. 3. The system validates the credentials. 4. The system authenticates the user and grants access according to their assigned role.
Alternative flows	A1 (step 2): The user cancels; they remain logged out.
Exception flows	E1 (step 3): Credentials are missing; the system identifies the required fields and allows correction. E2 (step 3): Credentials are incorrect; the system rejects login and allows another attempt. E3 (step 4): Login cannot be completed; the system reports failure and the user remains logged out.
Business rules	Each user has exactly one role: Student or Teacher.
Non-functional constraints	Use HTTPS; never expose plaintext passwords. Support keyboard navigation, labeled fields, and identifiable validation errors.

7.3 Use Case UC-04

Use case ID	UC-04
Use case name	Manage courses
Brief description	A teacher creates, renames, or deletes their courses.
Primary actor	Teacher
Secondary actors	None
Priority	Must
Related requirements	FR-02A, FR-02B
Trigger	The teacher selects a course management action.
Pre-conditions	The teacher is logged in. Their course list contains only courses they own.
Post-conditions	The course is created, renamed, or deleted as requested.
Main flow	1. The teacher selects creating a course. 2. The system displays the course creation form. 3. The teacher enters a course name and submits. 4. The system validates the name. 5. The system creates the course with the teacher as its owner and confirms success.
Alternative flows	A1 (step 1): Rename. The teacher selects a course, enters a new name, and submits. The system validates and saves the name, then confirms success. A2 (step 1): Delete. The teacher selects a course for deletion. The system requests confirmation. The teacher confirms, and the system deletes the course and confirms success. A3: The teacher cancels before submitting changes or confirming deletion; no changes are made.
Exception flows	E1 (step 4 / A1): The name is empty; the system requests a name and allows correction. E2 (step 5 / A1 / A2): The operation fails; the system reports failure without confirming success.
Business rules	A course name is mandatory. Each course belongs to exactly one teacher. Teachers can view and manage only their own courses. Deletion requires confirmation.
Non-functional constraints	Enforce authentication and ownership checks on the server. Denied actions must not change course data.

7.4 Use Case UC-05

Use case ID	UC-05
Use case name	Manage course materials
Brief description	A teacher publishes or deletes materials in their course.
Primary actor	Teacher
Secondary actors	None
Priority	Must
Related requirements	FR-04A
Trigger	The teacher selects a material management action.
Pre-conditions	The teacher is logged in and has opened a course they own.
Post-conditions	The material is published or deleted as requested.
Main flow	1. The teacher selects publishing a material. 2. The system displays the material form. 3. The teacher provides the material and submits. 4. The system validates the material. 5. The system publishes it, makes it accessible to enrolled students, and confirms success.
Alternative flows	A1 (step 1): Delete. The teacher selects an existing material and requests deletion. The system deletes it and confirms success. A2: The teacher cancels before submitting; no material is published.
Exception flows	E1 (step 4): The material is empty or invalid; the system identifies the error and allows correction. E2 (step 4): An uploaded file exceeds 20 MiB or has a disallowed type; the system rejects it and allows replacement. E3 (step 5 / A1): Publishing or deletion fails; the system reports failure without confirming success.
Business rules	Only the course owner can publish or delete materials. Enrolled students can access published materials. Uploaded files must be non-empty, at most 20 MiB, and of an allowed type.
Non-functional constraints	Enforce authentication, course ownership, and upload validation on the server. Rejected uploads must not be published.

7.5 Use Case UC-06

Use case ID	UC-06
Use case name	Join a course
Brief description	A student requests to join a course using an invite code or link.
Primary actor	Student
Secondary actors	None
Priority	Must
Related requirements	FR-03C
Trigger	The student enters an invite code or opens an invite link.
Pre-conditions	The student is logged in.
Post-conditions	The student's request appears in the course roster's pending list. The student remains unenrolled until approved by the teacher.
Main flow	1. The student submits an invite code. 2. The system validates the code and identifies the course. 3. The system checks that the student is neither enrolled nor already pending. 4. The system adds the request to the roster's pending list and confirms that teacher approval is required.
Alternative flows	A1 (step 1): The student opens an invite link; the system validates its invitation and continues at step 3.
Exception flows	E1 (step 2 / A1): The invitation is invalid; the system informs the student and creates no request. E2 (step 3): The student is already enrolled or pending; the system informs them and creates no duplicate request. E3 (step 4): Saving fails; the system reports failure without confirming the request.
Business rules	Only students can request enrollment. Teacher approval is required before enrollment and access to course content. Duplicate requests and enrollments are prohibited.
Non-functional constraints	Enforce authentication and student-role checks on the server. Pending students must not gain access to protected course content.

7.6 Use Case UC-13

Use case ID	UC-13
Use case name	Submit an assignment
Brief description	A student submits work for a course assignment.
Primary actor	Student
Secondary actors	None
Priority	Must
Related requirements	FR-05B, FR-05C
Trigger	The student selects an assignment for submission.
Pre-conditions	The student is logged in and enrolled in the course. The assignment exists.
Post-conditions	The submission is saved with its submission time and marked late if submitted after the deadline.
Main flow	1. The student opens the assignment. 2. The system displays its details and submission status. 3. The student uploads a file and submits. 4. The system checks whether submission is allowed and validates the file. 5. The system saves the submission, records its time, marks it late if applicable, and confirms success.
Alternative flows	A1 (step 2): Remove submission. The student requests removal of their existing submission. The system checks whether changes are allowed, removes it, and allows a new submission through step 3. A2 (step 3): The student cancels; no submission is saved.
Exception flows	E1 (step 4 / A1): The deadline has passed and late submissions are disabled; the system denies submission or removal and preserves any existing submission. E2 (step 4): The file is empty, exceeds 20 MiB, or has a disallowed type; the system rejects it and allows replacement. E3 (step 4): A submission already exists; the system requests its removal before accepting a new one. E4 (step 5 / A1): Saving or removal fails; the system reports failure without confirming success.
Business rules	An existing submission must be removed before submitting again. After the deadline, submission and removal are allowed only if the teacher permits late submissions. New submissions after the deadline are marked

	late. Uploaded files must be non-empty, at most 20 MiB, and of an allowed type.
Non-functional constraints	Enforce authentication, enrollment, deadline settings, and file validation on the server. Confirm submission success only after both the file and submission record are persistently stored.

8 Prototype / Mockups

Screen	Related use case(s)	Mockup / link
Login	UC-01, UC-02	login
My courses	UC-03, UC-04, UC-06	my-courses
Course	UC-05, UC-07, *UC-15, UC-19*	course/:id
Course Participants	UC-08, UC-09, UC-10	roster
Course Assignments	UC-11, UC-12, UC-13, *UC-14*	assignments
Course Discussions	*UC-16*	forum
My profile	*UC-17*	my-profile
My notifications	*UC-18*	notifications

9 Requirements Traceability Matrix

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
FR-01A	UC-01	Login		
FR-01B	UC-02	Login		
FR-01C	UC-01–UC-19 (role restrictions as applicable)	All screens (role restrictions as applicable)		
FR-02A	UC-03, UC-04	My courses		
FR-02B	UC-03, UC-04	My courses		
FR-03A	UC-08, UC-09, UC-10	Course Participants		
FR-03B	UC-08, UC-10	Course Participants		
FR-03C	UC-03, UC-06, UC-08, UC-09	My courses, Course Participants		
FR-04A	UC-05	Course		
FR-04B	UC-07	Course		
FR-05A	UC-11, UC-12	Course Assignments		
FR-05B	UC-11, UC-13	Course Assignments		
FR-05C	UC-13	Course Assignments		

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
FR-06A	UC-14	Course Assignments		
FR-07A	UC-15, UC-19	Course		
FR-07B	UC-16	Course Discussions		
FR-08A	UC-18	My notifications		
FR-08B	Not specified in Section 6	<Not decided yet>		
FR-09A	UC-17	My profile		
FR-09B	Not specified in Section 6	<Not decided yet>		

10 Requirements Review

#	Issue found	Requirement(s)	Resolution

11 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Trần Hoàng Minh Khang	ChatGPT	Suggest rules for Section 5.3 (BR-01 to BR-08)	Reviewed with Huỳnh Trung Ngạn; shortened the rules and removed extra details to match the requirements in Section 5.1 and 5.2.
2	Trần Hoàng Minh Khang	ChatGPT	Suggest wireframe layouts for Section 8 (Prototype / Mockups)	Redrew the wireframes in Figma for both Teacher and Student views, connected interactive prototype flows, and checked all links with Huỳnh Trung Ngạn.
3	Huỳnh Trung Ngạn	ChatGPT, Perplexity	FIll in content for Section 2, 3, 4	Verified against common business rules and written contract of PA1 with Perplexity
4	Huỳnh Trung Ngạn	ChatGPT	FIll in content for Section 9	Verified if the content was actually consistent with requirements and wireframes